Overload as a Cusp Catastrophe: Identifiability Limits and a Ground-Truth Failure of Early-Warning Indicators in Wearable Physiology

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Abstract—Overload is usually modelled by naming states and counting the transitions between them. Such models fit anything and forbid nothing. We derive it instead from the geometry of a cusp catastrophe: a latent load coordinate relaxing in a quartic potential whose tilt follows momentary demand and whose bistability follows accumulated recovery debt. Functional states become metastable basins, and the transition kernel follows from Kramers escape rates. The geometry fixes two exponents before any data are seen. Hysteresis width grows as the 3/2 power of the splitting factor; near the fold, variance and log-autocorrelation scale as $\mp 1/2$ in distance to the tipping point. Estimation is exact, since conditional on the slow variable the drift is linear in a reparameterised coefficient vector. Three findings follow, of which two concern method rather than physiology. A Monte Carlo study over 1.2×10^4 simulated series shows that at the lengths wearable corpora supply, only the relaxation rate and the diffusion coefficient are recovered. The parameters carrying the geometry are ratios with an unbounded denominator, so length buys their ordering but never their value: rank correlation for the splitting factor reaches 0.88 at $n = 1000$ while its linear correlation stays at 0.10. That bounds any cusp analysis of such data, this one included. Next, the rolling-window variance and autocorrelation indicators the early-warning literature runs on return exponents of the wrong sign on data simulated from a model that does contain a fold, so we withdraw our own early-warning test rather than report it. Finally, across 147 recordings from 40 people in three open corpora, the one identifiable prediction is not supported above a surrogate-calibrated null; measured power bounds this to excluding strong and moderate dynamics. The work is motivated by autistic adults. None are analysed here, and a pre-registered validation is specified.

Index Terms—Critical transitions, cusp catastrophe, hysteresis, early-warning signals, identifiability, electrodermal activity, wearable sensing, sensory overload.

I. Introduction

Autistic adults describe functional breakdown under combined sensory and cognitive demand in consistent terms: it arrives suddenly, and it clears slowly [1]. Adult cohort studies couple sensory atypicality to executive difficulty with large effect sizes [2], [3], so the phenomenon is not in doubt. What is missing is a formal account of the dynamics.

Existing models do not supply one. Computational work on autism is dominated by Bayesian and predictive-coding framings whose empirical support a ten-year review found mixed and methodologically inconsistent [4], and the most recent dynamical treatment of symptom trajectories is avowedly theoretical, collecting no data [5]. Applied state-transition work assigns states from experimental condition labels and counts a transition matrix. That is close to circular: the recovered matrix mostly re-describes the block structure of the protocol, and yields nothing observation could contradict.

We take a different route. Rather than positing states and fitting transitions between them, we posit a one-dimensional stochastic relaxation in a cusp potential and let the phenomenology follow. The load coordinate x_t sits in $V(x; a, b) = \frac{1}{4}x^4 - \frac{1}{2}ax^2 - bx$, where the normal factor b_t carries momentary demand and the splitting factor a_t carries depleted regulatory capacity. The substantive claim is that capacity does not add to load. It changes the geometry. Below a critical a the load tracks demand and returns; above it a second attractor appears, bringing a fold, a hysteresis loop and critical slowing down. Sudden onset and slow recovery stop being observations to report and become consequences to check — numerical exponents that data can reject, since the geometry is closed-form. This paper is about how far that promise survives contact with real wearable data.

Contributions. (i) A cusp formulation in which the five functional states are geometric regions rather than labels, and the 5×5 kernel is generated by nine interpretable parameters instead of twenty counted frequencies (Sec. II). (ii) An exact profiled-least-squares estimator, with a high-replication identifiability study bounding what any such analysis can claim, ours included (Sec. IV). (iii) A ground-truth demonstration that the standard early-warning indicators report the wrong sign on data containing a genuine fold, and a measurement of the eightfold size inflation of the nominal test on near-unit-root series (Sec. V). (iv) A calibrated null on 147 recordings, with power characterised so the result is a bound rather than an absence.

The target population is autistic adults. No public wearable dataset of that group exists at the temporal density this model needs [6], [7], so none are analysed here — and we say so rather than title the work after a group it does not measure.

II. The Cusp–Hysteretic Model

A. Fast dynamics and a slow debt variable

The latent load obeys a gradient relaxation with additive noise,

$$\mathrm{d}x_t = \lambda\big[-x_t^3 + a_t x_t + b_t\big]\mathrm{d}t + \sigma\,\mathrm{d}W_t, \tag{1}$$

with control parameters linked to observed drivers and to a slow variable A_t that accumulates recovery debt:

$$b_t = \beta_0 + \beta_S S_t + \beta_T T_t + \beta_U U_t, \quad a_t = \alpha_0 + \alpha_A A_t, \tag{2}$$

$$\mathrm{d}A_t = \varepsilon[\mathrm{sp}(x_t - x_{\mathrm{ref}}) - A_t]\mathrm{d}t, \qquad \varepsilon \ll 1. \tag{3}$$

Here sp is a softplus, S_t, T_t and U_t are sensory, task-switch and uncertainty drivers, and λ sets the timescale without touching the geometry. Load above baseline accrues debt, debt raises a_t, and a higher a_t widens the hysteresis loop, so one episode makes the next easier. That is a mechanistic reading of the refractory period qualitative accounts describe.

B. Folds and two scaling laws

Equilibria solve $x^3 - ax - b = 0$, whose discriminant $4a^3 - 27b^2$ is positive exactly when two stable basins coexist. Folds sit where $V''(x) = 0$, at $x_\pm = \pm\sqrt{a/3}$, giving critical drives $\pm b_c(a)$ with $b_c(a) = \frac{2}{3\sqrt{3}}a^{3/2}$.

Proposition 1 (Hysteresis width, P1). For $a > 0$ the system jumps to the upper branch at $b = +b_c$ and returns at $b = -b_c$, so the loop has width $\Delta b(a) = \frac{4}{3\sqrt{3}}\,a^{3/2}$.

Linearising (1) about a stable equilibrium gives a relaxation rate $\lambda_{\mathrm{rel}} = 3x^{*2} - a$ that vanishes at a fold, so critical slowing down is a theorem here rather than an assumption. With $\mu = b_c - b$ the distance to the fold, the normal form gives $\lambda_{\mathrm{rel}} \propto \mu^{1/2}$, hence

Proposition 2 (Early-warning exponents, P2). $\mathrm{Var}(x) \propto \mu^{-1/2}$ and $-\log \mathrm{AC1} \propto \mu^{+1/2}$.

These are point hypotheses about exponents, differing in kind from the usual directional claim that an indicator rises before a transition — which the sceptical literature shows arises spuriously on almost any autocorrelated series [8], [9].

C. States and transition kernel

Crossing basin membership with the bistability condition partitions the (a, b) plane into five regions (Fig. 1b): calm and focused in the lower basin when monostable, stuck in the lower basin while an overloaded attractor coexists, overloaded in the upper basin, and recovering there once demand has fallen below the level that would have put the system there. The last two matter most: qualitative work uses both words freely, and to our knowledge neither has previously had a precise mathematical meaning. Between-basin moves are noise-induced escapes with Kramers rate $r = \frac{1}{2\pi}\sqrt{|V''(x^*)V''(x_s)|}\exp(-2\Delta V/\sigma^2)$ [10], so the kernel is a deterministic function of (a_t, b_t, σ): twenty counted probabilities replaced by nine interpretable parameters that move with demand.

III. Materials and Methods

A. Corpora

Three open corpora, each taken from its primary repository rather than a mirror, span a gradient of experimental control: WESAD, 15 subjects in a controlled laboratory protocol [11]; Wearable Exam Stress, 10 students across 30 examinations [12]; and Nurse Stress, 15 nurses across roughly 1250 h of hospital shifts [13]. All supply Empatica E4 electrodermal activity, blood volume pulse, skin temperature and accelerometry. Windowing and quality screening leave 147 recordings from 40 people, median 194 windows each.

B. Sensor assignment and a negative control

Read the load coordinate and its drivers off the same sensor and the model stops being falsifiable, since shared measurement noise alone manufactures the coupling; we therefore assign them to disjoint sensors. Which signal carries the load is settled by measurement, not preference: the coordinate of a slow relaxation process must itself be slow. At a 30 s window tonic electrodermal activity has lag-1 autocorrelation 0.98, while a cardiac index from heart rate and RMSSD has 0.19, RMSSD over so few beats being dominated by beat-detection error. Configuration B (tonic EDA) is therefore primary and C (skin temperature) a robustness check. We keep A, the cardiac index, as a negative control: a working pipeline should find nothing in it.

C. Estimation

Taking the Euler map of (1) as the model, rather than as an approximation to it, makes the transition density exactly Gaussian and removes discretisation bias. Expanding the drift, the mean increment is linear in a reparameterised coefficient vector,

$$\Delta x_t = \theta_1(-x_t^3) + \theta_2 x_t + \theta_3 A_t x_t + \theta_4 + \theta_5 S_t + \theta_6 T_t + \theta_7 U_t + \eta_t, \tag{4}$$

with $\theta_1 = \lambda$, $\theta_2 = \lambda\alpha_0$, $\theta_3 = \lambda\alpha_A$ and $\theta_{4:7} = \lambda\beta_{0:U}$. Since A_t depends only on ε, the fit is ordinary least squares profiled over one scalar: exact, no bounds, no starting values, no local optima. We impose $\theta_1 \geq 0$, since a negative value puts the system outside the model rather than merely fitting it badly, and we keep units where that constraint binds so no reported proportion is conditioned on its own outcome.

D. Nulls

The nominal t-test on θ_1 is unusable here. Tonic EDA is close to a unit-root process, and regressing on an integrated series inflates significance in the classical way. We calibrate every test against surrogate ensembles: random-walk surrogates matched on length, increment variance and marginal scale, and amplitude-adjusted Fourier surrogates (IAAFT [14]) that preserve spectrum and marginal, so only nonlinearity can drive a rejection. Competing models — homogeneous Markov, covariate logistic, Gaussian

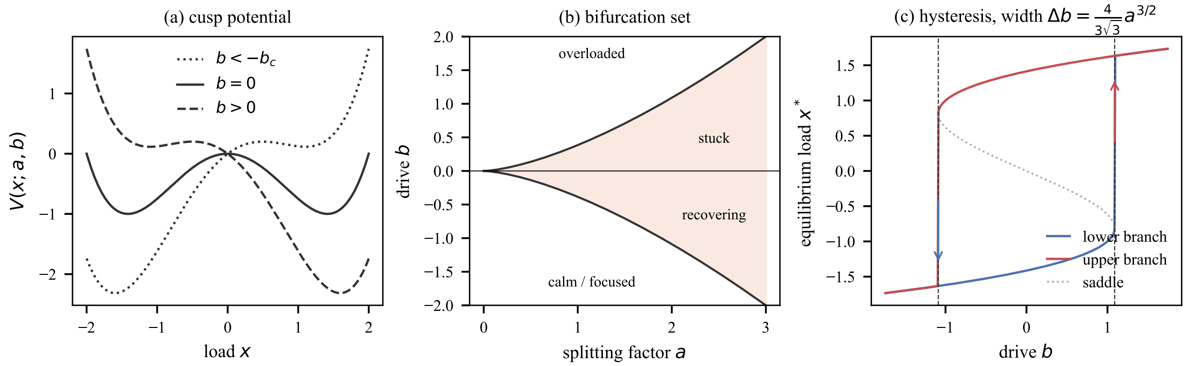

Fig. 1. The geometry from which everything else follows. (a) The cusp potential at three levels of drive b. (b) The bifurcation set in the (a, b) plane, with the five functional states appearing as regions rather than as assigned labels. (c) The hysteresis loop, whose width is fixed by Prop. 1 rather than merely being positive.

HMM, the nested Ornstein–Uhlenbeck case and gradient boosting — are scored by held-out one-step predictive log-density.

IV. What This Data Can Identify

Before testing anything we asked which parameters are recoverable at the series lengths on offer. This is not a formality. It turned out to be the result that organises the rest of the paper.

We simulated from the model with randomised true parameters and refitted, 2000 replicates at each of six lengths, and compared true against estimated values (Table I, Fig. 2a).

Two correlations are reported, because they answer different questions and the gap between them is the finding. Spearman's ρ asks whether the ordering survives; Pearson's r asks whether the number can be read off. For λ and σ they agree, and both are high from $n = 100$ up. Everywhere else they diverge.

The splitting factor carries the geometry, so it is the case that matters. Its rank recovery climbs with length, $\rho = 0.33$ at $n = 100$ to 0.88 at $n = 1000$. Its value recovery does not climb at all: $r = 0.15$ at $n = 200$, $r = 0.10$ at $n = 1000$. There $\hat{\alpha}_0$ spans $[-140, 3.3]$ against a 99th percentile of 2.0, so one blow-up in two thousand replicates destroys the linear correlation. That is the model's own arithmetic, not a quirk of the simulation.

The consequence is sharp, and it binds any study of this kind rather than only ours. Recovered geometry is built from the ratios $a = \theta_2/\theta_1$ and $b = \theta_4/\theta_1$, and nothing bounds θ_1 away from zero. Both therefore inflate without bound as $\hat{\lambda}$ falls (Fig. 2b). That explains the pattern in Table I exactly: dividing by a small number preserves order, which is why ρ recovers, and destroys scale, which is why r does not.

The real fits behave the same way. Among units where the constraint does not bind, the median $\hat{\lambda}$ is 0.0059 and the median recovered $\hat{\alpha}_0$ is 12 — attractors near ± 2 on a coordinate standardised to unit variance. These are not weakly bistable systems. They are systems with no

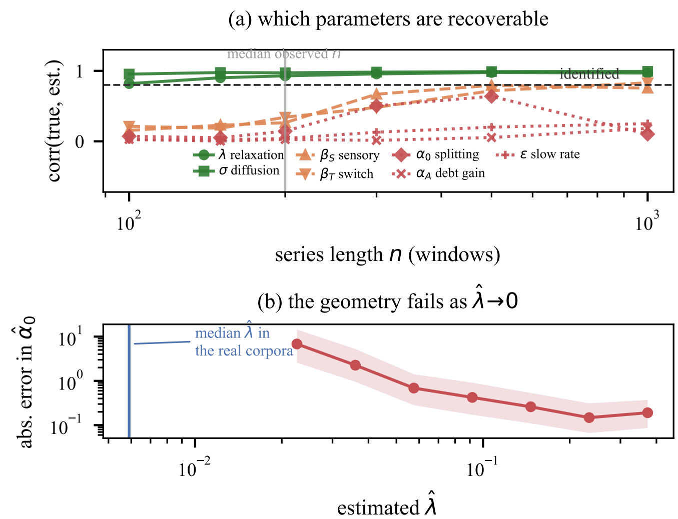

Fig. 2. (a) Rank recovery against series length, 2000 replicates per point. (b) The mechanism: error in the recovered splitting factor rises as $\hat{\lambda}$ falls, because $a = \theta_2/\theta_1$. The median $\hat{\lambda}$ in the real corpora lies below the whole simulated range.

TABLE I

Parameter recovery, 2000 replicates per cell. ρ (Spearman) asks whether the ordering is recovered, r (Pearson) whether the value is. Only λ and σ score well on both. For α_0 they diverge and stay diverged.

	Spearman ρ		Pearson r	
Parameter	n=200	n=1000	n=200	n=1000
λ (relaxation)	**0.94**	**0.98**	**0.93**	**0.97**
σ (diffusion)	**0.99**	**1.00**	**0.97**	**0.99**
β_S (sensory drive)	0.64	0.91	0.26	0.76
β_T (task switch)	0.58	0.88	0.34	0.83
α_0 (splitting)	0.57	0.88	0.15	0.10
α_A (debt gain)	0.32	0.72	0.03	0.18
ε (slow rate)	0.10	0.44	0.04	0.25

restoring cubic term whatever, whose apparent geometry is an artefact of the division. That median $\hat{\lambda}$ also falls below the entire range our simulation covers, so the recordings sit further into the non-identified regime than any of the 1.2×10^4 synthetic series used to map it.

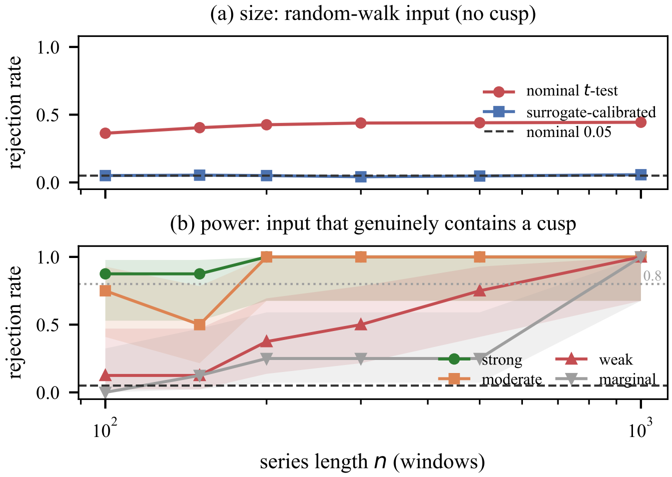

Fig. 3. (a) Size on random-walk input: the nominal test rejects at roughly 0.42 at every length, surrogate calibration restores nominal size. (b) Power on input that genuinely contains a cusp. Bands are Wilson intervals.

The practical reading is narrower than "collect more data": length buys a defensible ranking by α_0, never a reportable value.

One of the five pre-stated predictions survives as evaluable here: the existence of the restoring cubic term, governed by θ_1, the only parameter recovered on both measures. P1 needs the ratio of two badly identified quantities and P2 the distance to the fold, while P3 through P5 rest on ε and α_A. We report that structure rather than four numbers dressed up as tests.

V. Results

A. The nominal test is invalid; the calibrated one is powered

Over 30,000 random-walk replicates spanning six lengths, the nominal one-sided t-test on θ_1 rejects 41.9% of the time rather than 5%. The inflation does not shrink with sample size. It grows, from 36.3% (95% CI $[35.0, 37.7]$) at $n = 100$ to 44.4% ($[43.0, 45.8]$) at $n = 1000$ — the signature of spurious regression on an integrated series, not a small-sample artefact. Surrogate calibration restores nominal size across the same grid: 5.0% pooled over 7,500 replicates, every per-length interval covering 0.05. Calibration costs power, so we quote power for the test actually used. At $n = 200$ it is 1.00 for strong cusp dynamics, 0.76 for moderate and 0.24 for weak, reaching 1.00 everywhere by $n = 1000$ (Fig. 3). The null below therefore excludes strong and moderate dynamics. It does not exclude a faint fold.

B. The early-warning estimator fails on ground truth

Proposition 2 describes the system. Testing it needs an estimator, and we checked ours before trusting it. On data simulated from a model that genuinely contains a fold, the exponent computed from the true equilibrium curvature comes back at $+0.565$ against a predicted $+0.5$: theory and implementation agree. The rolling-window indicators the early-warning literature relies on [15], [16] do not. Over a grid of window lengths, detrending choices and series lengths reaching $n = 20{,}000$, rolling $-\log \mathrm{AC1}$ returns negative slopes where $+0.5$ is predicted, and rolling variance returns positive slopes where -0.5 is (Table II). The sign is wrong in every configuration tested.

TABLE II
Recovering the exponent from ground-truth cusp data. The closed-form theory does; the rolling-window estimators return the wrong sign.

Estimator	Predicted	Detrend	Window	Median
theory $\lambda_{\mathrm{rel}}(\mu)$	$+0.5$	—	—	$\mathbf{+0.565}$
$-\log$AC1	$+0.5$	no	30	-0.262
$-\log$AC1	$+0.5$	no	120	-0.035
variance	-0.5	no	30	$+0.311$
variance	-0.5	yes	120	$+0.036$

This is structural, not noise. Critical slowing down means the relaxation time diverges at the fold, so a fixed-width window saturates exactly where the effect is largest; detrending then strips the low-frequency power carrying what survives. An order of magnitude more data does not recover the sign. We withdraw our own P2 test as uninterpretable, and the caution generalises: a directional early-warning result obtained this way is weak evidence either way.

C. The identifiable prediction is not supported

Under the calibrated test the restoring cubic term is significant in 2.7% of the 147 units, against a nominal 5%: 0.0% for WESAD ($n = 15$), 0.0% for the examinations ($n = 30$), 3.9% for the nurse sessions ($n = 102$). Because those 102 units come from 15 nurses we also weight one vote per person, giving 1.5% (Wilson interval $[0.0, 3.1]$%), with 10% of people showing at least one significant session. That last figure is the most generous reading available and is still what chance produces across several sessions each.

Held-out prediction agrees from another direction. The full model does not beat its own nested monostable special case: it wins on 33.8% of units, median difference -0.25 nats per step, Wilcoxon $p = 8{\times}10^{-5}$ in the simpler model's favour. The monostable model is this model with the cubic term switched off, so that is the cleanest available statement that the term is doing no work. Prospective onset prediction gives AUC between 0.50 and 0.46 at lead times of 3 to 12 windows. Chance.

D. Structure without fold structure

The null is specific, and calling it noise would be wrong. Two of the three calibrated statistics do exceed their surrogate nulls: the likelihood ratio against the monostable restriction at 40.8% of units, and marginal bimodality at 49.7%. Only the statistic isolating the cubic term sits at chance. A bimodal marginal with no cubic drift is precisely the configuration in which a study reporting distributional evidence alone concludes the opposite of the truth.

Robustness is mixed, and we report it that way. Configuration B sits at nominal under both the random-walk (0.046) and the harder IAAFT (0.069) ensembles, and holds across 30 s to 120 s windows. A and C are elevated, at 0.11 to 0.16. Three explanations were tested and all three failed: IAAFT did not bring C down, clipping rates run opposite to the effect, and an AR(1) size study (ϕ from 0 to 1) found no persistence at which the calibrated test over-rejects. What is left is nonlinear structure the surrogates cannot reproduce. But A is the negative control, built on a coordinate already shown to be noise-dominated, so the parsimonious reading is motion and peak-detection artefact rather than a fold. A test that fires on the negative control tells us significance does not mean cusp there, which is why the conclusion rests on B.

VI. Discussion and Conclusion

Modelling overload as motion in a cusp potential trades a descriptive account for a generative one whose consequences compute in advance. States become basins, the kernel a function of nine parameters, sudden onset and delayed recovery point hypotheses. Precision of that kind costs something: the model can lose. Here it did.

The identifiability result is the more durable contribution. Every attempt to fit cusp geometry to wearable physiology inherits the same ratio structure, and the remedy depends on what is wanted. Ranking participants by α_0 needs length, order $n \gtrsim 500$ windows. Reporting a value needs a reparameterisation estimating a and b directly, since length alone never fixed it in our simulations. A splitting factor of 12 from a 200-window recording is not a finding, and the arithmetic producing it is not specific to our implementation.

Limitations. No autistic participants are analysed, so the motivating claim is untested rather than refuted. Tonic electrodermal activity is a peripheral, artefact-prone proxy for a latent construct. Weak cusp dynamics stay compatible with everything reported here. Our estimator critique covers fixed-width rolling indicators with and without Gaussian detrending, which is what the applied literature mostly runs; it says nothing about adaptive-bandwidth or spectral estimators, which we did not test. Note also that longer records rescue rankings, not values, and not every parameter even in rank: the slow rate ε reaches only $\rho = 0.44$ at $n = 1000$.

Toward validation. We pre-specify the study that would test the mechanism in its intended population: 28 days of ecological momentary assessment with continuous wrist physiology in autistic adults and a matched comparison group, using the same estimator, nulls and falsification criteria as here. The primary hypothesis is a between-group difference in α_0, better powered than the within-group detection attempted here, with $n \gtrsim 500$ windows per participant so that parameter sits inside the identifiable regime. The debt mechanism needs more than length: with α_A and ε unrecoverable from passive data at any duration tested, P3 to P5 will need either experimental control over recovery intervals or a hierarchical fit that pools them across participants. A second well-powered null under those conditions would be evidence that overload is not a cusp catastrophe at physiological timescales.

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